Carmichael function

In number theory, a branch of mathematics, the **Carmichael function** $\lambda(n)$ of a positive integer n is the smallest positive integer m such that

$$a^m \equiv 1 \pmod{n}$$

holds for every integer a coprime to n. In algebraic terms, $\lambda(n)$ is the exponent of the multiplicative group of integers modulo n. As this is a finite abelian group, there must exist an element whose order equals the exponent, $\lambda(n)$. Such an element is called a **primitive λ-root modulo *n***.